

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

University Examination May 2010

Semester – II B.Tech. (CL/EC/ME)/

Semester – I B.Tech. (CE/EE/IT)

ME-102 : Fundamentals of Mechanical Engineering

Date: 14.05.10, Friday

Time: 12:30 p.m. to 3:30 p.m.

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Make suitable assumptions, usual notations and draw neat figures wherever if required.

SECTION-I

- Q:1 (a). What is difference between power & energy? Give their SI units. [02]
(b). What is pressure? How is it measured? Differentiate between gauge pressure & absolute pressure. [03]
(c). Explain combine gas law with P-V diagram. What is characteristic gas constant & explain its significance. [05]

- Q: 2 (a). For adiabatic process, prove that : $\left(\frac{T_2}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{(r-1)}{r}} = \left(\frac{V_1}{V_2}\right)^{(r-1)}$ [05]
(b). 0.72 Kg of gas at 14 bar and 290⁰ C is expanded to four times the original volume according to the law $PV^{1.3} = \text{constant}$. Calculate (i) The original and final volume of the gas (ii) the final temperature of the gas (iii) the final pressure of the gas. Take $R=287 \text{ J/kg K}$. [05]

OR

- Q: 2 (a). What is clearance volume? Why is it provided in reciprocating compressor? Differentiate between reciprocating & rotary compressor. [05]
(b). Calculate the dimensions(diameter & stroke) of double acting reciprocating compressor neglecting clearance volume from following a data: Compressor power=10 kw, compressor ratio=7, Inlet pressure=1 bar, piston speed=180 m/min., Law of compression= $pV^{1.3} = \text{constant}$. $L/D = 1.5$ [05]
Q: 3 (a). Explain the working of four stroke petrol engine with neat diagram. [05]
(b). An oil engine works on diesel cycle with temperature of 25⁰ C at starting of compression. If the ratio of adiabatic compression is 16 and that of adiabatic expansion is 9. Find the efficiency of cycle. Take $\gamma=1.4$ for air. [05]
(c). What is the difference between coupling & clutch? With a simple sketch explain the working of Oldham's coupling. [05]

OR

- Q:3 (a). A four cylinder two stroke cycle petrol engine develops 30 KW at 2500 RPM. The indicated mean pressure of each is 800 KPa and $\eta_{\text{mech}} = 80 \%$. Calculate the diameter and stroke of each cylinder if $L/D=1.5$. Also, calculate the brake specific fuel consumption of the engine if brake thermal efficiency is 28 %. The calorific value of petrol is 44000 KJ/Kg. [05]
(b). What are the elements of power transmission? Classify the mechanical drives. [05]
(c). Write brief notes on: (i) Types of gear (ii) Types of gear trains. [05]

SECTION-II

- Q:4 (a). Write the advantages of non-conventional energy sources over conventional sources of energy. [02]
- (b). Explain: (i) wet steam (ii) dry saturated steam and (iii) superheated steam [03]
- (c). Explain latent heat of evaporation. Calculate the internal energy of wet steam having pressure 12 bar and dryness fraction = 0.85. [05]

- Q:5 (a). Differentiate between lower calorific value & higher calorific value of fuel. [05]
Describe in brief about Biogas as fuel with advantages.
- (b). Draw the neat sketch of Babcock & Wilcox boiler and explain its working. [05]
- (c). The following data were obtained with a separating & throttling calorimeter. [05]
Steam pressure in main pipe = 10 bar
Water separated in separating calorimeter = 0.4 kg.
Steam discharge from throttling calorimeter = 5 kg.
Manometer reading = 170 mm of Hg
Barometer reading = 760 mm of Hg
Temperature of steam after throttling = 130°C
Determine dryness fraction of steam. $C_{ps} = 2.1 \text{ KJ/Kg K}$.

OR

- Q:5 (a). What is boiler draught? Write down the function of boiler draught. [05]
Explain in brief: (i) Induced draught (ii) Forced draught (iii) Balanced draught
- (b). Enlist the different accessories of boiler. Draw neat sketch of Economiser and state its function, location & advantages. [05]
- (c). A boiler generates 2700 kg of steam per hour at pressure of 10 bar from feed water at 40°C . The coal consumption rate of boiler is 300 kg/hr. The calorific value of coal is 30000 KJ/Kg. The dryness fraction of steam is 0.82
Calculate: (i) Heat supplied per hour by fuel (ii) Equivalent evaporation per kg of coal (ii) Boiler efficiency [05]

- Q:6 (a). Explain the working principles of (i) Heat engine (ii) Heat pump (iii) Refrigerator with neat sketch. [05]
- (b). What is the difference of operating principle between positive displacement pump and roto-dynamic pump? Describe the working of the double acting reciprocating pump with neat sketch. [05]

OR

- Q:6 (a). Differentiate between refrigeration & air conditioning. Draw a neat sketch of Split air conditioner & state its advantages. [05]
- (b). What is priming & self-priming of pump? Why priming is required in centrifugal pump? [05]